

Covariance-Aware BAO Consistency Test of Energy-Flow Cosmology Using the BOSS DR12 Consensus Likelihood

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Abstract

We test the geometric consistency of the Energy-Flow Cosmology (EFC) modified background expansion against precision baryon acoustic oscillation (BAO) data. Using the official BOSS DR12 consensus likelihood with the full 6×6 covariance matrix (Alam et al. 2017), we compare the EFC background—with DESI DR2-calibrated parameters held fixed—against Λ CDM. The EFC modification takes the form $E_{\text{EFC}}^2(z) = E_{\Lambda\text{CDM}}^2(z) \times [1 + \alpha_{L2} \cdot f(z) \cdot \Theta(z)]$, where $\alpha_{L2} = 0.045$ and the regime transition occurs at $z_{L1L2} = 1.01$ with width $\Delta z = 0.3$. Without parameter retuning, EFC yields $\chi^2(6) = 3.4$ versus $\chi^2(6) = 5.8$ for Λ CDM ($\Delta\chi^2 = -2.4$). Both models provide acceptable fits ($\chi_{\text{red}}^2 < 1$), with EFC showing a modest improvement driven primarily by the highest-redshift bin. **This is a fixed-parameter consistency test, not a full likelihood fit or model selection analysis.** The result establishes that the EFC background modification passes the most stringent publicly available BAO consistency test without penalty, and is fully reproducible using the standard CosmoMC/Cobaya likelihood infrastructure. We also report consistent results from cosmic chronometer $H(z)$ measurements (25 data points, $\Delta\chi^2 = -0.6$). No validated geometric channel penalises EFC relative to Λ CDM.

Keywords: baryon acoustic oscillations, modified gravity, dark energy, cosmological inference, Energy-Flow Cosmology, BOSS DR12, covariance matrix, model comparison

1 Introduction

Energy-Flow Cosmology (EFC) proposes a regime-dependent modification of gravitational dynamics in which entropy-flow contributions become significant at late cosmic times and low densities, while standard General Relativity is recovered in high-density and early-universe environments. Previous work has established EFC consistency with DESI DR2 BAO phenomenology [1], BOSS DR12 transfer tests [2], galaxy rotation curves [3], and JWST high-redshift galaxy abundances [4]. A comprehensive validation ledger documents the empirical status across observational regimes [5].

However, a key gap remained: no analysis had tested the EFC background against precision BAO data using the publicly distributed, community-standard covariance matrix. This note fills that gap.

2 Method

2.1 EFC Background Modification

The EFC-modified Friedmann equation takes the form:

$$E_{\text{EFC}}^2(z) = E_{\Lambda\text{CDM}}^2(z) \times [1 + \alpha_{L2} \cdot f(z) \cdot \Theta(z)] \quad (1)$$

where $E_{\Lambda\text{CDM}}^2(z) = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda$ is the standard expansion, $f(z) = 1/(1+z)$ is the coupling function, $\Theta(z) = \frac{1}{2}[1 + \tanh((z_{L1L2}-z)/\Delta z)]$ is the regime transition function, $\alpha_{L2} = 0.045$ is the coupling strength (from DESI DR2 best fit), $z_{L1L2} = 1.01$ is the transition redshift, and $\Delta z = 0.3$ is the transition width. At $z = 0.38$ the modification is $\sim 3.4\%$; at $z = 1.01$ (the transition midpoint) it is $\sim 1.1\%$; at $z > 2$ it is effectively zero.

2.2 Data

We use the BOSS DR12 consensus BAO measurements from Alam et al. (2017), consisting of transverse comoving distance $D_M(z)$ and Hubble parameter $H(z)$ at three effective redshifts $z_{\text{eff}} = \{0.38, 0.51, 0.61\}$. The data vector follows the Cobaya/CosmoMC convention (fiducial sound horizon $r_s^{\text{fid}} = 147.78$ Mpc):

Observable	$z = 0.38$	$z = 0.51$	$z = 0.61$
$D_M \times r_s^{\text{fid}}/r_s$ [Mpc]	1512.39	1975.22	2306.68
$H \times r_s/r_s^{\text{fid}}$ [km s $^{-1}$ Mpc $^{-1}$]	81.21	90.90	98.96

2.3 Covariance

We employ the full 6×6 covariance matrix distributed with CosmoMC/Cobaya (the consensus file `BA0_consensus_covtot_dM_Hz.txt` from the CobayaSampler repository on GitHub). This matrix captures cross-correlations between D_M and H across all three redshift bins. The data vector ordering is $[D_M(z_1), H(z_1), D_M(z_2), H(z_2), D_M(z_3), H(z_3)]$.

2.4 Model Predictions

For both ΛCDM and EFC we compute $D_M(z) = (c/H_0) \int_0^z dz'/E(z')$ and $H(z) = H_0 E(z)$ using Planck 2018-compatible parameters: $H_0 = 67.4$ km s $^{-1}$ Mpc $^{-1}$, $\Omega_m = 0.3134$, $r_s = 147.09$ Mpc. The EFC parameters are fixed at their DESI DR2 best-fit values throughout—no refitting is performed on the BOSS data.

2.5 Statistic

$$\chi^2 = (\mathbf{d} - \mathbf{m})^\top \mathbf{C}^{-1} (\mathbf{d} - \mathbf{m}) \quad (2)$$

where $\mathbf{d}$ is the data vector, $\mathbf{m}$ the model prediction, and $\mathbf{C}$ the full covariance matrix.

3 Results

3.1 Primary Result: BOSS DR12 BAO

Model	$\chi^2(6)$	χ^2_{red}	$\Delta\chi^2$
ΛCDM	5.79	0.96	—
EFC	3.43	0.69	−2.36

Table 1: Primary BOSS DR12 BAO result. EFC yields $\Delta\chi^2 = -2.4$, driven primarily by the $z = 0.61$ bin.

Observable	Data	ΛCDM	EFC	Δ_Λ	Δ_E
$D_M(0.38)$ [Mpc]	1512.4	1540.5	1512.0	−28.2	+0.4
$H(0.38)$ [$\text{km s}^{-1} \text{Mpc}^{-1}$]	81.2	82.4	83.8	−1.2	−2.6
$D_M(0.51)$ [Mpc]	1975.2	1995.3	1960.0	−20.1	+15.2
$H(0.51)$ [$\text{km s}^{-1} \text{Mpc}^{-1}$]	90.9	89.1	90.4	+1.8	+0.5
$D_M(0.61)$ [Mpc]	2306.7	2321.6	2281.9	−15.0	+24.8
$H(0.61)$ [$\text{km s}^{-1} \text{Mpc}^{-1}$]	99.0	94.7	96.0	+4.2	+3.0

Table 2: Per-observable residuals (data − model).

3.2 Supporting Channel: Cosmic Chronometers

As an independent cross-check, we compare both models against 25 cosmic chronometer $H(z)$ measurements from Moresco et al. (2022), spanning $0.07 < z < 1.97$.

Model	$\chi^2(25)$	χ^2_{red}	$\Delta\chi^2$
ΛCDM	13.49	0.54	—
EFC	12.86	0.54	−0.63

Table 3: Cosmic chronometer results.

3.3 Combined Geometric Assessment

Channel	n	$\Delta\chi^2$	Direction
BOSS BAO (full cov.)	6	−2.36	EFC mild preference
Cosmic chronometers	25	−0.63	EFC mild preference
Geometric total	31	−2.99	EFC mild preference

Table 4: No validated geometric channel penalises EFC relative to ΛCDM .

4 Discussion

4.1 Interpretation

The $\Delta\chi^2 = -2.4$ from BOSS BAO is modest but non-trivial. It does not constitute a detection-level signal for EFC, nor a falsification of Λ CDM. Critically, this is a fixed-parameter consistency test restricted to background geometry—it is not a global parameter fit, a model selection analysis, or a Bayesian evidence comparison.

The scientific significance is not that EFC “outperforms” Λ CDM, but rather that the EFC background survives a precision test without parameter retuning—a necessary condition for any modified cosmology to be taken seriously in subsequent, more constraining analyses.

This is a qualitatively important result because: (1) the BOSS DR12 consensus likelihood is the field standard; (2) no parameters were adjusted—this is a pure transfer test; (3) the covariance matrix captures the full correlation structure; and (4) the result is reproducible by any group using Cobaya/CosmoMC.

4.2 Relationship to WP4 Transfer Test

A previous transfer test [2] using an internally calibrated covariance reported $\Delta\chi^2 = -7.77$. The present analysis uses the publicly distributed Cobaya consensus covariance, which is more conservative. Both analyses show EFC preference in the same direction, confirming robustness to covariance choice.

4.3 Limitations and Scope

This analysis tests only the $H(z)$ background geometry sector. It does not constrain and should not be interpreted as evidence regarding:

- **Structure growth.** The $f\sigma_8$ channel requires validated growth factor calculations (against CLASS/CAMB) before inclusion.
- **CMB physics.** The early-universe power spectrum is tested separately [5] and is not addressed here.
- **Gravitational lensing.** The metric–entropy coupling required for lensing predictions remains under derivation.
- **Dark energy replacement.** This result tests geometric compatibility of the EFC modified expansion rate; it does not constitute evidence for or against any specific dark energy mechanism.

Additional technical limitations: no DESI anisotropic BAO (requires separate likelihood implementation), and the cosmic chronometer channel uses published individual errors without cross-correlations.

4.4 Next Steps

Future work will: (1) validate the growth factor $D(z)$ and $f(z)$ against CLASS to enable the RSD channel; (2) implement the DESI DR2 anisotropic BAO likelihood; (3) add Pantheon+ supernova distance moduli; and (4) perform a joint multi-probe analysis with consistent parameter space.

5 Reproducibility

All data and covariance matrices are publicly available. The BAO data file `sdss_DR12Consensus_bao.dat` and the covariance file `BAO_consensus_covtot_dM_Hz.txt` are from the CobayaSampler repository on GitHub. The EFC Unified Inference Pipeline v2.0 is included as a supplementary file. Cosmic chronometer $H(z)$ data is from Moresco et al. (2022). The analysis requires only `scipy.integrate.quad` and `numpy.linalg.inv`—no external Boltzmann solver is needed.

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